

Moidul Hasan

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Research Interests

- AI in healthcare
- Biomedical signal & image processing
- mHealth- and IoT-based healthcare solutions
- Computer vision
- Clinical decision support systems
- Biophysics simulation

Education

Bangladesh University of Engineering and Technology
Bachelor of Science in Biomedical Engineering

Feb 2020 – Mar 2025
CGPA: 3.49/4.0

Standardized Test Performance

Test	GRE				IELTS				
Section Score	Quant	Verbal	Total	AWA	Listening	Reading	Writing	Speaking	Overall
	170	153	323	3.5	9.0	8.5	7.0	6.5	8.0

Relevant Coursework

- Medical Imaging
- Biomedical Signals and Systems
- Biomedical Instrumentation and Measurements
- Bioinformatics
- Digital Signal Processing
- Embedded Systems and Interfacing

Research Experience

Research Engineer, mHealth Research Group
Principal Investigator (PI): Prof. Taufiq Hasan, BME, BUET

Mar 2025 – Present

RadAssist

- Collaborated with the Institute of Epidemiology, Disease Control and Research (IEDCR) to develop a tuberculosis detection tool for a local dataset
- Improved Shenzhen, Montgomery, DA, and DB tuberculosis CXR datasets' interpretability with annotation box generation with the help of local radiologists

HEV Study

- Developed RIG system for HEV kit image data collection with a homogeneous setup for low-cost Bangladeshi conditions for Institute for Developing Science and Health Initiatives (ideSHi)
- Implemented AI algorithm for enhanced HEV detection from HEV test kit data in collaboration with International Centre for Diarrhoeal Disease Research, Bangladesh (icddr,b)

Undergraduate Research Assistant, mHealth Research Group
Supervisor: Prof. Taufiq Hasan, BME, BUET

Jan 2024 – Mar 2025

Undergraduate Thesis: Deep Learning Models for Tuberculosis Detection from Chest X-rays

- Localized TB regions with weakly supervised explainable AI methods
- Improved model focus using knowledge distillation for better interpretability
- Enhanced generalization across datasets with interpretable predictions

Projects

Undergraduate Capstone Project : View-Retina: Low-cost Smartphone Ophthalmoscope
Supervisor: Asst. Prof. Shams Nafisa Ali, BME, BUET

Jul 2022 – Nov 2023

- Designed a low-cost, portable smartphone ophthalmoscope to enhance retinal screening accessibility
- Developed novel image processing method for early glaucoma detection by calculating cup-to-disc ratio (CDR)
- Utilized adaptive thresholding and histogram smoothing for accurate optic disc and cup segmentation

- Implemented a deep learning multiclass classification model for *referable diabetic retinopathy detection*
- Developed a novel web application platform to facilitate retinal image diagnosis

- Co-Founder & Chief Research Officer (CRO); led feasibility study analyzing ground data
- Developing an AI-assisted agro-drone to reduce pesticide waste and chemical exposure via aerial spraying
- Raised BDT 80K in pre-seed funding from the World Bank-funded University Innovation Hub Program Cohort

Publications

Journal Paper

Shawkat, M.A., **Hasan, M.**, & Hasan, T., "Weakly Supervised Tuberculosis Localization in Chest X-rays through Knowledge Distillation", (**Manuscript in Preparation**)

Poster Presentation

Shawkat, M.A., **Hasan, M.**, & Hasan, T., "Knowledge Distillation Improves Tuberculosis Feature Learning in CNN", **BEAR Summit & Bangladesh National Semiconductor Symposium, Dhaka, Bangladesh, 2025**

Achievements

2024	University Innovation Hub Program (UIHP) Cohort Winner (BUET)
2024	Healthcare Case Competition Winner, BUET BME Fest 2.0
2022	Bangladesh Biology Olympiad (BDBO) Best Enzyme Award, Dhaka South

Technical Skills

Programming Languages	C/C++, Python, MATLAB, Arduino
Deep Learning Frameworks	TensorFlow, PyTorch
Python Libraries	NumPy, Pandas, Matplotlib, SciPy, Scikit-learn
CAD & Design	AutoCAD, SolidWorks, MakerBot, PreForm
Simulation & Modeling	Ansys Fluent, COMSOL Multiphysics, Simulink, Unity Game Engine
Productivity & Collaboration	Microsoft 365, Google Workspace, LaTeX, Canva, Origin

Internship

GETWELL Pharma Limited, Pran-RFL Group

May 2023

- Developed a comprehensive understanding of syringe manufacturing, assembly, and ISO-compliant packaging.
- Acquired knowledge of ethylene oxide sterilization protocols and quality control testing methodologies.
- Gained exposure to pharmaceutical production and analytical instrumentation (HPLC, FTIR).

Leadership Skills

Bangladesh Biology Olympiad Committee, Dhaka South, Assistant Organizing Secretary

Dec 2021 - Mar 2025

Planned and executed regional and national biology Olympiads, developing educational content and outreach initiatives to promote biological sciences among local school and college students.

BUET Biomedical Engineering Society (BMES) Student Chapter, Vice President

Dec 2021 - Mar 2025

Contributed to society development through editorial leadership. Edited and published the inaugural issue of *Bioluminescence*, featuring alumni interviews and academic highlights.

BUET Animata, Co-Founder & Vice President (Communications)

Nov 2023 - Mar 2025

Founded Bangladesh's first university-based animation and digital arts club. Led sponsor engagement, digital strategy, and member recruitment efforts to expand the club's academic and creative footprint.

BUET ACS Student Chapter, Event Management Chair

Jan 2024 - Mar 2025

Managed new member onboarding and event logistics. Organized *ChemBridge: Advancing Science Together*, the chapter's annual flagship symposium connecting students and professionals in chemical sciences.